



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
Summer Term Examination
Academic Year: 2024-25

Class: SY	Subject: Manufacturing Processes - I	Subject Code: PCC-ME206/ PCC-ME209
Date: 23.06.2025	Time: 14:30 to 17:30	Max. Marks: 100

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/ Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.	Solve all the questions	Marks	CO	BT
Q.1	Suggest the manufacturing process for the following component and justify your answer i) Spanner ii) Engine Block iii) Rail Wheel	15	CO4	BT1, BT2
Q.2	Explain the different types of components and terminology in the sand-casting process	15	CO2	BT1
Q.3	Classify the welding Process in detail and explain briefly with a diagram	15	CO3	BT2
Q.4	Explain in detail the Tube making, Explosive welding, and Stretch forming	15	CO4	BT2
Q.5	A rectangular block of steel C.S. 100 x 150 mm and 250 mm height is to be cast without any riser. The block is moulded entirely in the drag of a green sand flask and is top gated. The cope of the flask is 200 mm height and the height of the metal during pouring is 100 mm above the cope level. A tapered sprue is employed and the gating ratio is 1:4:2. Assume no energy losses in the system. The time taken (in seconds) to fill the casting cavity, if the tapered sprue with 500 mm exit area is used.	10	CO2	BT3
Q.6	Determine the die and punch size for both punching and blanking a circular disc of 20 mm diameter from a C20 steel sheet whose thickness is 1.5 mm. Also, calculate the Punching force and stripping force required in the operation. Consider the shear strength for C20 is 294 MPa	10	CO5	BT3
Q.7	An arc welding DC power source has a linear power source characteristic with open circuit voltage=80V and short circuit current is 1000 Amps, the voltage arc length characteristic of the arc is given by $20+4L=V$, where L-length of arc in mm. i) The optimum length of the arc for obtaining maximum arc power at welding is ii) The voltage and current settings should be done on the power source for maximum arc power is	10	CO3	BT3
Q.8	The casting shown (Fig. 1) is to be made in CI using a wooden pattern. Assuming shrinkage allowance and machine allowance, calculate the dimensions of the pattern. All dimensions are in cm, and the thickness is 3 cm	10	CO2	BT3

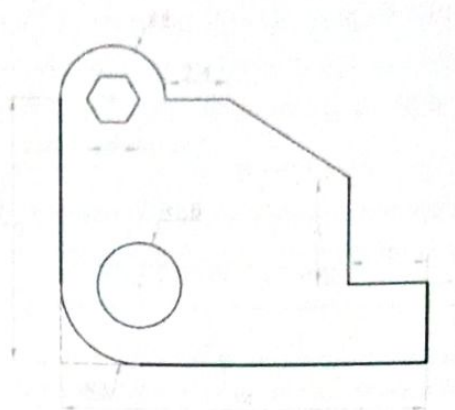


Fig :1

Data for Problem

Shrinkage allowance		
Material	Dimension	Shrinkage allowance (inch/ft)
Cast Iron	Up to 2 feet	0.125
	2 feet to 4 feet	0.105
	over 4 feet	0.083
Machining allowance		
Material	Dimension (inch)	Allowance (inch)
Cast Iron	Up to 12	0.12
	12 to 20	0.20
	20 to 40	0.25

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Categorize and explain various manufacturing processes, understanding their role and application in industrial production.
CO 2	Design and analyse casting processes, including mould making, pattern making, and identification of casting defects.
CO 3	Develop a thorough understanding of different welding techniques, their operational principles, and potential defects.
CO 4	Acquire the skills to describe and apply the principles of metal forming operations in manufacturing processes.
CO 5	Explain and perform sheet metal forming operations



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
Summer Term End Examination
Academic Year: 2024-25

Class: SY	Subject: Materials Science / Engineering	Subject Code: PCC-ME203 / PCC-ME205
Date: 24.06.2025	Time: 14:30 to 17:30	Max. Marks: 100

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q1	Select the material for the given component, and justify your answer with a scientific reason (mechanical properties, structure, microstructure, etc of the material) a) Lath bed c) Wind turbine blade b) Hex saw blade d) Furnace Wall	20	CO1 CO2	BT3
Q.2	What are the different types of methods to manufacture the powder	15	CO4	BT2
Q.3	Explain the types of composite material in detail with a diagram	15	CO1	BT1
Q.4	Elaborate on the iron-iron carbon diagram in detail, along with their reactions and microstructure.	15	CO2	BT1
Q.5	What is bulk heat treatment? explain their types in detail with a temperature vs time chart and their microstructure	15	CO3	BT1
Q.6	Write down the 2 advanced materials, explain with the properties, advantages and applications	10	CO5	BT1
Q.7	Is there any use of TTT diagram? If yes, then illustrate with a diagram of the microstructure	10	CO4	BT1

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Identify the various materials, viz., metals, polymers, ceramics, and composites
CO 2	Analyse the dependence of mechanical properties on the microstructure
CO 3	Select appropriate heat treatment for metals and alloys for particular application
CO 4	Acquire knowledge about powder metallurgy process and its applications.
CO 5	Acquire knowledge about advanced materials and its applications.

	SGGS Institute of Engineering and Technology, Vishnupuri, Nanded		
	Academic Year: 2024-25	Semester: II	Examination: Summer Term
	Class: Second Year	Branch: Mechanical Engineering	
	Date: 20/06/2025	Time: 14:30 - 17:30	
Subject: Thermodynamics		Code: PCC-ME201/ME202	Maximum Marks: 100

General Instructions:

- I. This paper contains 2 (Two) pages. Figures to right indicate full marks.
- II. Draw neat & compact sketches wherever necessary.
- III. Assume data wherever required and state them clearly.

Que. No.	Question	Marks	CO	BT
1.	Explain following situation from <i>first and second law of thermodynamics</i> perspective: Consider two persons Khagesh and Chakrsh. Khagesh has 10 friends who always stays together during parties and pass time. However, they seems busy when Khagesh needs them. On other hand Chakresh has five friends and are never too busy for him and are available when Chakresh needs them. <i>Who has more friends?</i>	10	CO2	BT2
2.	Explain following cycles in brief: a. Ideal Otto cycle. b. Ideal Brayton Cycle	10	CO5	BT1
3.	Along with the mathematical expression, explain in details the <i>increase of entropy principle</i> .	10	CO4	BT2
4.	A piston-cylinder device contains 0.05 m^3 of a gas initially at 200 kPa. At this state, a linear spring that has a spring constant of 150 kN/m is touching the piston but exerting no force on it. Now heat is transferred to the gas, causing the piston to rise and to compress the spring until the volume inside the cylinder doubles. If the cross-sectional area of the piston is 0.25 m^2 , determine: a. Final pressure inside the cylinder. b. Total work done by the gas. c. Fraction of this work done against the spring to compress it.	5 + 5 + 10	CO1	BT4
5.	Reversible heat engine operates between two reservoirs at temperature of 600°C and 40°C . The engine drives a reversible refrigerator which operates between reservoir at temperature of 40°C and -20°C . The heat transfer to the heat engine is 20000 kJ and the net work output of combined engine refrigerator is 360 kJ. a. Evaluate the heat transfer to the refrigerator and the net heat transfer to the reservoir at 40°C . b. Reconsider (a) given that maximum efficiency of heat engine and the cop of the refrigerator are each 40% of their maximum possible values.	10 + 10	CO3	BT5
6.	A hypothetical device is supplied with 2 kg/s of air at 4 bar, 300 K. Two separate streams of air leaves the device at ambient pressure of 1 bar. with equally divided mass flow rates. One of the exit stream is at 330 K while other is at 270 K. Determine whether such a device is possible.	10	CO4	BT4

Que. No.	Question	Marks	CO	BT
7.	Diesel Engine has a compression ratio (r_k) of 14, and cut-off takes place at 6% of the stroke. Find the air standard efficiency, given as $\eta = 1 - \frac{1}{\gamma \times r_k^{\gamma-1}} \times \frac{r_c^{\gamma-1} - 1}{r_c - 1},$ where r_k and r_c is cut-off and compression ratio respectively and γ is ratio of specific heat for air.	10	CO5	BT5
8.	A reversible heat engine operates between two reservoirs at temperature of 600°C and 40°C. The engine drives a reversible refrigerator which operates between reservoirs at temperature of 40°C and -20°C. The heat transfer to the heat engine is 20000 kJ and the net work output of the combined engine refrigerator plant is 360 kJ. Evaluate the heat transfer to the refrigerant and the net heat transfer to the reservoir at 40°C.	10	CO2	BT5

**** ALL THE BEST ****

CO1: To impart the principles of work and energy.

CO2: To inculcate fundamentals of thermodynamics laws, concepts & principles.

CO3: To comprehend knowledge about properties of pure substance.

CO4: To comprehend knowledge about entropy, irreversibility and availability

CO5: To make familiar with various thermodynamic cycles and to apply thermodynamic concepts in various applications.

BT1: Remember, **BT2:** Understand, **BT3:** Apply, **BT4:** Analyze, **BT5:** Evaluate, **BT6:** Create.